Prior Research for Global Shipping Disruption Analytics

1. Context and background

Global trade moves mostly by sea, and a small number of maritime chokepoints carry a large share of it. Recent research gives us three groups of work that are relevant to our project.

The first group is about **measuring trade from vessel data**. Arslanalp, Koepke, and Verschuur (2021) built daily port and trade indicators for Pacific island countries from satellite vessel tracking (AIS). They improved the cargo estimates, checked port calls against liner schedules, and presented the indicators as a complement to official statistics for spotting turning points and shocks. A later IMF study (Arslanalp et al., 2026) used PortWatch vessel data to estimate monthly country-level trade and compared the results with official data for Brazil, Jamaica, Japan, Samoa, and the United States. It found a better fit for advanced economies and large emerging markets. This supports using shipping indicators, but it also shows that they should be checked against independent data.

The second group is about **chokepoint closures and spillovers**. Pratson (2023) combined mapped shipping lanes with trade data to model what happens if any of 11 chokepoints is closed, and how trade would shift to other chokepoints and countries. Verschuur, Lumma, and Hall (2025) modeled hazards at chokepoints and estimated yearly trade exposure and economic losses from delays, rerouting, insurance, and freight costs.

The third group is about **rerouting and recovery**. Devetak, Verschuur, and Klimek (2026) used an agent-based model of 35,954 ships serving 1,651 ports. They found that a route's exposure alone does not predict the losses that actually happen. Rerouting can reduce losses at some ports while moving delays to other places, and losses grow with the length of the closure.

2. Research gap

The sources we reviewed cover AIS-based measurement, modeled closure impacts, and rerouting mechanisms. But we did not find a single standardized comparison of crisis-related traffic decline and recovery time across chokepoints, using the same definitions for every event. Also, the closure studies are mostly scenario models, not an analysis of the day-by-day traffic that was actually observed.

3. Justification of our project and method

Our project addresses this gap in a focused way. We compare several real crises using a common baseline and common definitions, and we report both the size of the traffic drop and how long recovery took. We use PortWatch because it provides daily estimated transit calls and cargo volume for a large set of chokepoints, and presents the results in a dashboard that a logistics or executive manager can use for supply risk monitoring.

4. Avoiding duplication

We are not repeating the closure models or the economic loss models. Those need extra data and modeling (costs, insurance, trade flows) that we don't have. Our work is a descriptive, observed-data comparison, so it complements them.

5. Conceptual framework

From the literature we took three outcomes to measure for each event:

1. **Local disruption:** the change in daily transits and estimated cargo volume at the affected chokepoint, compared with a baseline from before the event.
2. **Duration and recovery:** the time from the start of the event until activity returns to the baseline, using a threshold that is fixed before comparing events.
3. **Network spillover:** changes at other chokepoints or alternative routes, where the data allows, to see rerouting.

6. Limitations

PortWatch values are estimates. A fall in traffic does not prove how much money was lost, so we report traffic and estimated volume, not economic loss. PortWatch also revises its methodology and coverage over time, so we record the download date and the methodology version, and we don't mix unreconciled versions.

References

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